

Department of the Navy | Office of Naval Research (ONR)

Technical Report Directive: ONR-TR-2026-BB67-MAT
Subject: Manufacturing Specification for Elastomer-Infiltrated Composite Metal Foams (E-CMF) as High-Energy Blast Mitigation Armor
Primary Materials: High-Purity Aluminum (Al), Titanium (Ti), and Cross-Linked Natural Rubber (NR) Elastomer Matrices.

1. Executive Summary & Design Paradigm

This report outlines the industrial production pipeline for **Elastomer-Infiltrated Composite Metal Foams (E-CMF)** for heavy hull shielding applications. By moving away from brittle, monolithic armor plates, this design relies on a dual-phase material system: a **rigid, open-cell metallic skeleton** that absorbs initial kinetic energy via micro-strut crushing, and an **interpenetrating natural rubber network** that dampens shockwaves and rebounds after deformation.

When scaled to a thick outer hull layer, this material creates a highly buoyant geometric profile—ideal for heavy-duty, protective marine assets.

2. Multi-Phase Material Selection & Properties

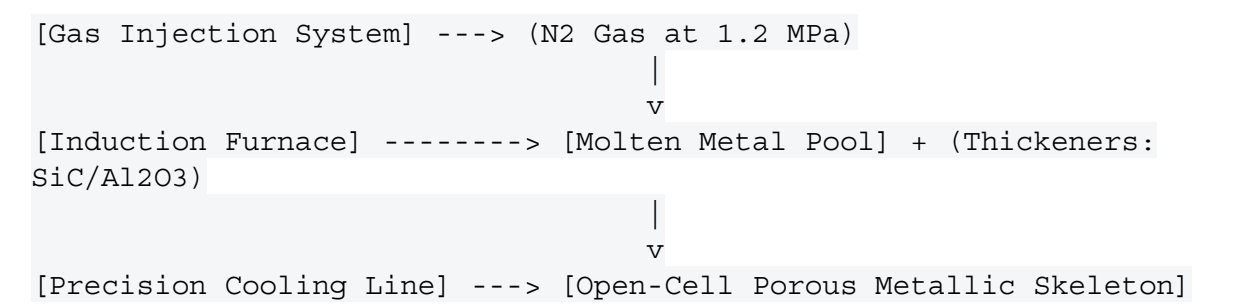
The metallic skeleton can be manufactured using either **Aluminum (Al-Alloy)** for lightweight, cost-effective energy damping or **Titanium (Ti-6Al-4V)** for extreme ballistic resistance. The infiltrating phase consists of a customized natural rubber formulation optimized for high-pressure injection.

Material Phase Component	Density (ρ , g/cm^3)	Tensile Strength (σ_{ts} , MPa)	Primary Structural Role
Aluminum Skeleton (Al-Mg Alloy)	2.70 (Solid) / 0.35 (Foamed)	220–310	Low-mass kinetic energy absorption via plastic cell deformation.
Titanium Skeleton (Ti-6Al-4V)	4.43 (Solid) / 0.72 (Foamed)	950–1050	High-velocity projectile fragmentation & shear mitigation.

Material Phase Component	Density (ρ , g/cm^3)	Tensile Strength (σ_{ts} , MPa)	Primary Structural Role
Natural Rubber Elastomer (NR)	0.92–0.95	25–35	Hyperelastic shock damping & cell rebound optimization.

3. Open-Cell Metallic Skeleton Manufacturing

The rigid skeleton is manufactured using a modified **Direct Melt Gas Injection** process using **Nitrogen (N_2)** as the primary blowing agent.



3.1 Slurry Thickening and Viscosity Control

Before gas injection, the molten metal pool must be stabilized to prevent the immediate collapse of the cell walls.

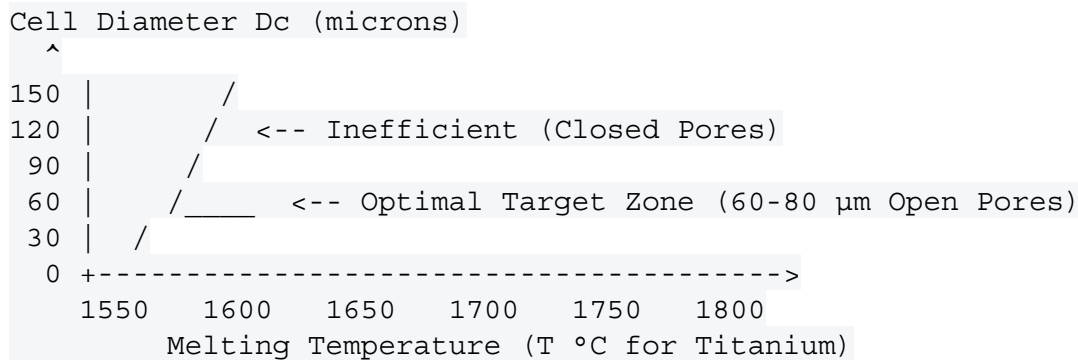
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- For **Aluminum (720°C)**: Add 15% Calcium (Ca) or Silicon Carbide (SiC) particles.
- For **Titanium (1700°C)**: Mix with 10% Aluminum Oxide (Al_2O_3) particles under vacuum induction melting to ensure uniform cell wall surface tension.
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3.2 Nitrogen (N_2) Gas Injection Dynamics

Nitrogen gas is introduced through an array of rotating ceramic impellers submerged at the bottom of the melt. The gas flow rate is regulated based on the required final porosity:
$$\Phi = 1 - \left(\frac{\rho_{\text{foam}}}{\rho_{\text{solid}}} \right)$$

The relationship between foaming temperature and resulting cell diameter (D_c) must be

monitored to ensure an optimal open-cell matrix:



4. Pressure-Assisted Elastomer Infiltration Pipeline

Once the open-cell metal foam has solidified and cooled, the residual skin is acid-etched to open any remaining surface pores. The block is then transferred to a high-pressure autoclave for **Polymer Infiltration Synthesis (PIS)**.

4.1 Natural Rubber Pre-Polymer Compound

The rubber infiltration matrix uses a low-viscosity liquid natural rubber pre-polymer:

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- **Base:** Cis-1,4-polyisoprene
- **Vulcanizing Agent:** 2.5% Elemental Sulfur (S_8)
- **Accelerator:** Zinc Oxide (ZnO) mixed with stearic acid.
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4.2 Vacuum-Pressure Infiltration Cycle

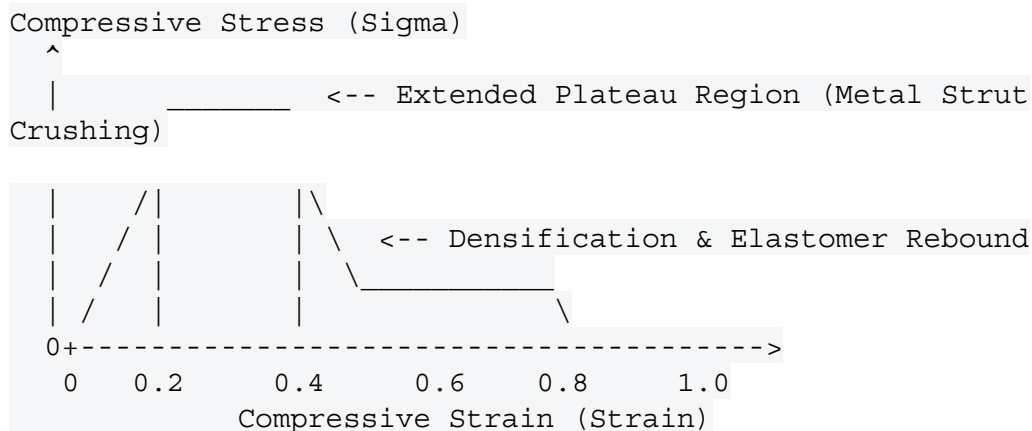
To eliminate internal air trapping and ensure an **Infiltration Efficiency $\geq 88\%$** , a four-step mechanical process is mandatory:

1. **Chamber Evacuation:** The open-cell metal skeleton is placed in the mold cavity and drawn down to a vacuum of $\leq 10^{-2}$ Torr for 45 minutes to evacuate all internal pore channels.
2. **Liquid Injection:** The liquid rubber mixture is introduced into the bottom of the mold under vacuum, completely submerging the metallic preform.
3. **Autoclave Overpressure:** The chamber is pressurized with dry Argon gas to **1.5 MPa**, forcing the elastomer deep into the micro-porosities of the metal cell walls.
4. **Thermal Vulcanization:** The internal temperature is elevated to **145°C** for 120 minutes while maintaining clamping force to cure the cross-linked rubber matrix

natively within the metal struts.

5. Mechanical Performance & Blast Response

The primary defensive advantage of E-CMF armor lies in its **Specific Energy Absorption (SEA)** under extreme dynamic loads.



The total energy absorbed per unit volume (W) is calculated during impact by integrating the stress-strain curve:

$$W = \int_0^{\epsilon_d} \sigma(\epsilon) \, d\epsilon$$

Where ϵ_d represents the densification strain limit where the internal rubber cells are completely compressed. The interpenetrating rubber network keeps the compressed metal struts confined, preventing the armor from cracking or falling away during secondary impacts.

6. Trusted Official Resources & References

1. **National Center for Biotechnology Information (NCBI / NIH):** *A Review of Different Manufacturing Methods of Metallic Foams* (PMCID: PMC10870358) — Process details on melt gas injection and gas bubbling thermodynamics.
2. **NASA Technical Reports Server (NTRS):** *Polymer Infused Composite Metal Foam as a Potential Aircraft Component* (Document ID: 20190031917) — Data on vacuum infiltration procedures, resin retention metrics, and 88% infusion efficiency standards.
3. **National Science Foundation (NSF) Public Access Repository:** *Robust Bicontinuous Elastomer–Metal Foam Composites* (Award Id: 10353325) — Production guidelines for pressure-assisted liquid rubber infiltration into open-cell metallic networks.
4. **Defense Technical Information Center (DTIC):** *Bio-inspired Armor Protective Material Systems for Ballistic and Blast Mitigation* (Accession Number: ADA543984) — Multi-phase structural performance under high dynamic blast impacts.
5. **Department of Transportation (PHMSA):** *Metal Foams for Impact Mitigation and Shock*

Absorption — Performance parameters detailing progressive cell wall deformation under heavy ballistic loads.